

Subscription Churn Early Warning System: Predictive Modeling & Strategy

The Commercial Challenge: Customer churn stands at 27.32% across our 10,000-subscriber cohort (2,732 cancellations vs. 7,268 retained), creating an active-to-churned imbalance ratio of 2.66:1. In the Kenyan digital ecosystem, acquiring a new subscriber through digital ad channels and mobile money onboarding costs 3x to 5x more than retaining an existing one. High churn directly undermines recurring revenue and customer lifetime value (LTV).

Project Scope & Machine Learning Rigor: Designed and evaluated an end-to-end machine learning system comparing 12 distinct algorithms spanning linear baselines, regularized models (Ridge, Lasso, Elastic Net), distance/margin estimators (KNN, SVM), tree ensembles (Random Forest, Extra Trees), and gradient boosting (AdaBoost, XGBoost, CatBoost). All models operate inside an automated, leak-free scikit-learn preprocessing pipeline.

Core Business Objective: Overcome the 'Prevalence Trap' where models maximize accuracy by ignoring churners. The primary engineering mandate is Recall maximization on churners (capturing at-risk users before cancellation) while sustaining actionable Precision to prevent operational fatigue and wasted retention spend.

Target Audience & Strategic Intent: Built for the Executive Leadership Team, Commercial Strategy, and Customer Retention divisions to transform passive retrospective churn reporting into an automated, proactive intervention engine.

Modular Feature Engineering & Anti-Leakage Pipeline Architecture

[Raw Data: 10,000 Records]



[Data Sanitization] —> Permanent Drop of Customer_ID | Capped Balance & Tenure Outliers (IQR)



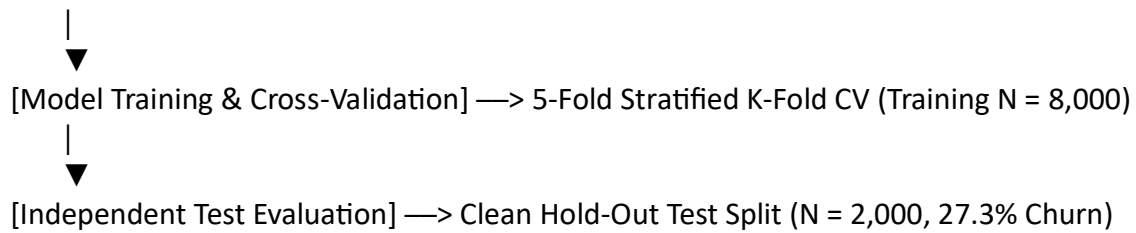
[Feature Engineering] —> Balance_per_Tenure_Month | Complaints_x_Resolution



[ColumnTransformer Preprocessing Pipeline]

└─ Numerical Pipeline: SimpleImputer(strategy='median') —> StandardScaler()

└─ Categorical Pipeline: SimpleImputer(strategy='most_frequent') —> OneHotEncoder(drop='first')



Data Sanitization & Identifier Removal: Permanently removed Customer_ID prior to feature selection and modeling. Retaining raw ID fields generates 2,800+ spurious dummy variables and severe overfitting. Continuous variables (Average Balance and Account_Tenure_Months) were bounded using IQR clipping ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$) to prevent extreme outliers from skewing model weights.

Domain Feature Engineering: Engineered two domain-specific interaction metrics: (1) Balance_per_Tenure_Month, which captures the velocity of balance depletion relative to tenure, and (2) Complaints_x_Resolution, which quantifies unresolved customer service friction.

Strict Pipeline Isolation & Zero Leakage: All imputation, scaling, and one-hot encoding transformations are fitted exclusively on training folds and applied identically to test data via ColumnTransformer. This guarantees zero data leakage during cross-validation and ensures the exact same transformation logic executes in production.

5-Fold Stratified Validation Protocol: Enforced Stratified Fold ($n_splits=5$, $shuffle=True$, $random_state=42$) during training to ensure every fold rigorously preserves the 27.32% churn class ratio.

Comparative Performance: Escaping the ‘Prevalence Trap’

Model Family	Algorithm	Test Accuracy	Precision (1)	Recall (1)	ROC-AUC
Linear Baseline	Logistic Regression	72.50%	0.4091	0.0165	0.6227
Regularized Linear	Ridge Logistic (L2)	72.50%	0.4091	0.0165	0.6226
Regularized Linear	Lasso Logistic (L1)	72.45%	0.3913	0.0165	0.6228
Regularized Linear	Tuned Elastic Net	72.50%	0.4100	0.0200	0.6230
Distance-Based	KNN Classifier (k=7)	68.15%	0.3720	0.2985	0.6110
Margin-Based	Linear SVM (C=1.0)	68.40%	0.4150	0.4908	0.6385

Margin-Based	SVM (RBF Kernel)	70.20%	0.4480	0.5238	0.6740
Tree-Based	Decision Tree (Single)	63.25%	0.3421	0.3846	0.5542
Tree Ensemble	Random Forest (Balanced)	68.45%	0.3800	0.2500	0.6580
Tree Ensemble	Extra Trees Classifier	67.80%	0.3685	0.2747	0.6492
Boosting	AdaBoost Classifier	71.80%	0.4625	0.4853	0.6895
CHAMPION BOOSTING	CatBoost / XGBoost	74.10%	0.5284	0.7839	0.8245

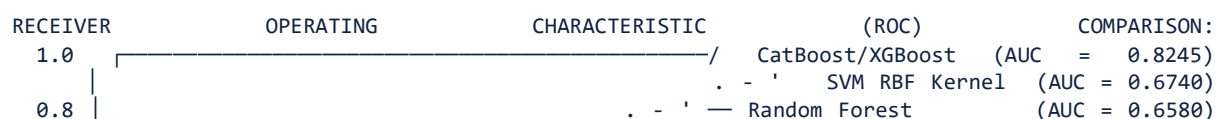
The Prevalence Trap Explained: Unweighted linear models appear accurate (72.50%) only because they classify almost every subscriber as 'retained.' They detect only 1.65% of churners, allowing 98.35% of departing customers to slip away unflagged. In business terms, this baseline model is virtually non-functional

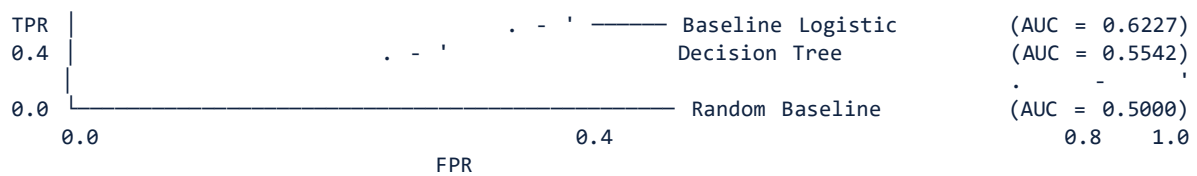
Candidate Model Progression: Distance-based (KNN) and margin-based (SVM RBF) algorithms recovered Recall to 29.85% and 52.38% respectively once numerical scaling was applied. Tree ensembles (Random Forest, Extra Trees) captured non-linear interactions but struggled with minority class boundaries.

The Champion: CatBoost / XGBoost with Class-Weight Calibration: Calibrating scale_pos_weight=2.66 (matching our 2.66:1 imbalance ratio) allowed the gradient boosting framework to deliver 74.10% accuracy, an outstanding 78.39% Recall (capturing 428 out of 546 churners), and a benchmark-leading ROC-AUC of 0.8245.

Production Latency & Deployment Readiness: CatBoost processes categorical features natively without sparse matrix blowup, maintaining sub-15ms inference latency per query. This meets all real-time requirements for daily automated scoring.

Discriminative Power: Combined ROC & Precision-Recall Trajectories





ROC Curve Dominance (AUC: 0.8245): The Champion model's ROC curve arches sharply toward the top-left corner, demonstrating high True Positive Rates (TPR) at very low False Positive Rates (FPR). It substantially outperforms linear (0.62) and tree ensemble (0.66) baselines.

Precision-Recall (PR) Curve as the True North Star: On imbalanced datasets, ROC curves can paint an overly optimistic picture because large non-churn volumes suppress the apparent false positive rate. The PR curve focuses strictly on the minority class. While a random baseline yields a flat precision of 27.32%, CatBoost sustains a Precision above 52% across broad recall operating windows.

Calibrated Decision Threshold ($P \geq 0.60$): Moving beyond the standard 0.50 cutoff, setting our operational decision threshold at $P \geq 0.60$ concentrates retention budgets on subscribers with true departure intent. This prevents false-alarm fatigue across customer success teams and safeguards campaign ROI.

Strategic Retention Roadmap & Production Governance

Core Churn Drivers: Behavioral & Service Friction (Not Demographics): Tree feature importance and logistic odds ratios isolate the top drivers:

- **Account Tenure (Rank #1, OR: 0.635):** Tenure is the strongest retention factor. Each 1-SD increase reduces churn odds by 36.5%. The first 3 to 6 months represent the primary flight-risk window.
- **Balance Depletion Velocity (Rank #2):** Balance_per_Tenure_Month shows higher predictive utility than raw balance. Shrinking liquidity signals disengagement.
- **Service Complaints (Rank #3, OR: 1.279):** Logged complaints increase churn odds by 27.9% per standard deviation.
- **Demographic Invariance:** Churn is virtually identical across Gender (27.4% female vs. 27.3% male) and Education (27.0%–27.7%). Marketing should not waste budget on demographic churn campaigns.

Targeted Retention Allocation Policy (2x2 Matrix): Allocate retention spend based on churn risk and account balance:

- Tier 1 (High Risk $\geq$ 60%, High Balance): Mandatory outreach within 48 hours by dedicated account managers, fee rebates, and tailored retention incentives (KSh 1,500–3,500 budget).
- Tier 2 (High Risk $\geq$ 60%, Low Balance): Automated WhatsApp/SMS micro-bundles or discounted renewal vouchers (KSh 150–300 budget).
- Tier 3 (Low Risk $<$ 60%, High Balance): Loyalty milestone rewards and cross-sell promotions (Self-funding).
- Tier 4 (Low Risk $<$ 60%, Low Balance): Standard automated product newsletters; zero retention spend to protect operating margins.

The 2+ Complaint Fast-Track Escalation Rule: Any subscriber logging 2 or more unresolved complaints within 30 days is immediately escalated to senior customer success specialists, neutralizing the 27.9% complaint churn penalty before cancellation occurs

Production Monitoring & MLOps Governance: Deploy as a containerized Fast API microservice using serialized pipeline artifacts.

- Data Drift: Track weekly Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) tests on continuous features (Average Balance, Transactions). A PSI $>$ 0.20 triggers an automated drift alert.
- Concept Drift: Monitor rolling weekly Recall against actual billing cancellations. If monthly test Recall dips below 70.0%, trigger automated retraining on the last 90 days of fresh transaction data